



TYWE2L Module Datasheet

Hardware Product Development > Network Modules > Wi-Fi Module >

WE Series Module

Version: 20210119

Contents

1 Overview	2
1.1 Features	2
1.2 Major application fields	3
2 Module interfaces	4
2.1 Dimensions and footprint	4
2.2 Pin definition	4
2.3 Test pin definition	5
3 Electrical characteristics	7
3.1 Absolute electrical characteristics	7
3.2 Electrical conditions	7
3.3 Wi-Fi TX power consumption	8
3.4 Wi-Fi RX power consumption	8
3.5 Power consumption in operating mode	9
4 RF features	11
4.1 Basic RF features	11
4.2 Wi-Fi output power	11
4.3 RF RX sensitivity	12
5 Antenna	13
5.1 Antenna type	13
5.2 Antenna interference reduction	13
6 Packaging information and poduction instructions	14
6.1 Mechanical dimensions	14
6.2 Production instructions	14
6.3 Recommended oven temperature profile	15
7 Appendix: Statement	16



TYWE2L is a low power-consuming built-in Wi-Fi module developed by Hangzhou Tuya Information Technology Co., Ltd. It consists of a highly integrated wireless radio chip (ESP8285) and several peripherals, with a built-in Wi-Fi network protocol stack and robust library functions.

1 Overview

TYWE2L also provides a low power-consuming 32-bit CPU, 1 MB flash, 50 KB SRAM, and various peripheral resources.

TYWE2L is an RTOS platform that integrates all the function libraries of the Wi-Fi MAC and TCP/IP protocols. Users can develop embedded Wi-Fi products as required based on the function libraries.

Figure 1 shows the schematic diagram of TYWE2L.

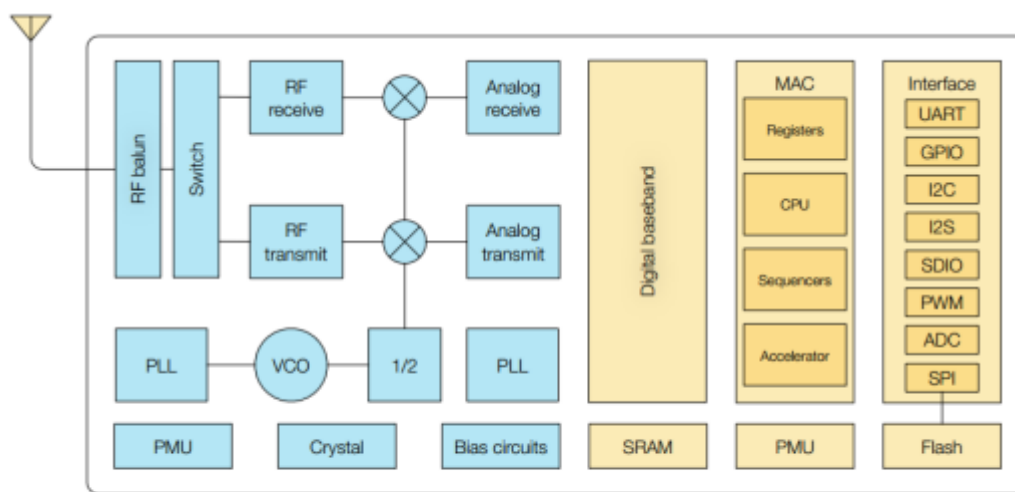


Figure 1 TYWE2L schematic diagram

Figure 1 TYWE2L schematic diagram

1.1 Features

Built-in low power-consuming 32-bit CPU, which can also be used as an application processor

- Basic frequency: 80 MHz and 160 MHz supported
- Working voltage: 3.0 V to 3.6 V
- Peripherals: five GPIOs
- Wi-Fi connectivity
 - 802.11 b/g/n

- Channel 1-14@2.4GHz(CH1-11 for US/CA, CH1-13 for EU/CN)
- WEP/WPA/WPA2/WPA2 PSK (AES) security mode supported
- Up to +20 dBm output power in 802.11b mode
- STA/AP/STA+AP working mode supported
- SmartConfig and AP network configuration modes supported (for Android and iOS devices)
- Onboard PCB antenna
- Working temperature: -20°C to +105°C

1.2 Major application fields

- Intelligent building
- Intelligent home and household applications
- Intelligent socket and light
- Industrial wireless control
- Baby monitor
- Network camera
- Intelligent bus

2 Module interfaces

2.1 Dimensions and footprint

TYWE2L provides two rows of pins with a distance of 2.0 mm between every two pins.

TYWE2L dimensions: 3 mm (W) x 15 mm (L) x 17.3 mm (H) (see figure 2)

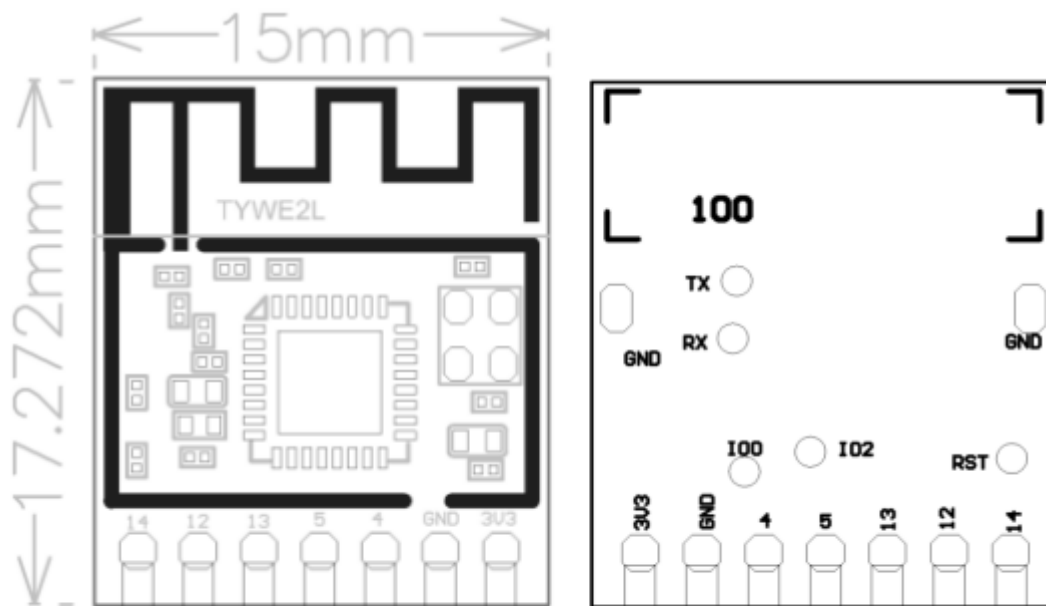


Figure 2 TYWE2L dimensions

2.2 Pin definition

Table 1 describes the interface pins.

Table 1 TYWE2L interface pins

No.	Symbol	I/O Type	Function
1	14	I/O	GPIO_14
2	12	I/O	GPIO_12

No.	Symbol	I/O Type	Function
3	13	I/O	GPIO_13
4	5	I/O	GPIO_05
5	4	I/O	GPIO_04
6	GND	P	Power supply reference ground pin
7	3V3	P	Module power supply pin (3.3 V)

Note: P indicates power-supply pins and I/O indicates input/output pins.

2.3 Test pin definition

Table 2 describes the test pins.

Table 2 TYWE2L test pins

No.	Symbol	I/O Type	Function
-	IO2	I/O	UART1_TXD (module information recording pin)
-	RST	I/O	Reset pin
-	IO0	I/O	IO0 is involved in module startup. If IO0 is disconnected, the module enters the normal running state.

Note: I/O indicates input/output pins. If IO0 is disconnected, the module enters the normal running state. If IO0 is at a low level, the module enters the firmware programming state. Test pins are not recommended.

3 Electrical characteristics

3.1 Absolute electrical characteristics

Table 3, Absolute electrical characteristics

Parameters	Description	Minimum value	Maximum value	Unit
Ts	Storage temperature	-20	105	°C
VCC	Power supply voltage	-0.3	3.6	V
Static electricity voltage (human model)	TAMB-25°C	-	2	KV
Static electricity voltage (machine model)	TAMB-25°C	-	0.5	KV

3.2 Electrical conditions

Table 4, Normal electrical conditions

Parameters	Description	Min	Typ	Max	Unit
Ta	Working temperature	-20	-	105	°C
VCC	Working voltage	3.0	3.3	3.6	V

Parameters	Description	Min	Typ	Max	Unit
VIL	I/O low-level input	-0.3	-	VCC*0.25	V
VIH	I/O high-level input	VCC*0.75	-	VCC	V
VOL	I/O low-level output	-	-	VCC*0.1	V
VOH	I/O high-level output	VCC*0.8	-	VCC	V
I _{max}	I/O drive current	-	-	12	mA

3.3 Wi-Fi TX power consumption

Table 5 TX power consumption during constant emission

Parameter	Mode	Rate	Tx power	Typ	Unit
IRF	11b	11 Mbit/s	+17 dBm	220	mA
IRF	11g	54 Mbit/s	+15 dBm	110	mA
IRF	11n	MCS7	+13 dBm	100	mA

3.4 Wi-Fi RX power consumption

Table 6 RX power consumption during constant emission

Parameter	Mode	Rate	Typ	Unit
IRF	11b	11 Mbit/s	76	mA
IRF	11g	54 Mbit/s	76	mA
IRF	11n	MCS7	76	mA

3.5 Power consumption in operating mode

Table 7 TYWE2L working current

Working Mode	Working Status(Ta=25°C)	Value	Max	Unit
EZ mode	The module is in EZ status and the Wi-Fi indicator quickly flashes.	80	415	mA
AP mode	The module is in AP status and the Wi-Fi indicator slowly flashes.	90	451	mA
Operation mode	The module is in connected status and the Wi-Fi indicator is steady on.	58.5	411	mA

Working Mode	Working Status(Ta=25°C)	Value	Max	Unit
Disconnection mode	The module is in disconnected status and the Wi-Fi indicator is steady off.	80	430	mA

4 RF features

4.1 Basic RF features

Table 8, Basic RF features

Parameter	Description
Frequency band	2.412-2.484 GHz
Wi-Fi standard	IEEE 802.11b/g/n (channel 1-14)
Data transmitting rate	11b: 1, 2, 5.5, 11 (Mbit/s)11g: 6, 9, 12, 18, 24, 36, 48, 54 (Mbit/s)11n: HT20 MCS0-MCS7
Antenna type	PCB antenna (default)

4.2 Wi-Fi output power

Table 9 TX power during constant emission

Parameter	Min	Typ	Max	Unit
RF average output power, 802.11b CCK mode	-	20	-	dBm
RF average output power, 802.11g OFDM mode	-	17	-	dBm

Parameter		Min	Typ	Max	Unit
RF average output power, 802.11n OFDM mode	MCS7	-	14	-	dBm
Frequency error		-10	-	10	ppm

4.3 RF RX sensitivity

Table 10 RX Sensitivity

Parameter		Min	Typ	Max	Unit
PER < 8%, RX sensitivity, 802.11b CCK mode	1 M	-	-91	-	dBm
PER < 10%, RX sensitivity, 802.11g OFDM mode	54 M	-	-75	-	dBm
PER < 10%, RX sensitivity, 802.11n OFDM mode	MCS7	-	-72	-	dBm

5 Antenna

5.1 Antenna type

TYWE2L uses the onboard PCB antenna.

5.2 Antenna interference reduction

To ensure optimal Wi-Fi performance when the Wi-Fi module uses the onboard PCB antenna, it is recommended that there be a space of at least 15 mm between the module antenna and other metal parts.

6 Packaging information and production instructions

6.1 Mechanical dimensions

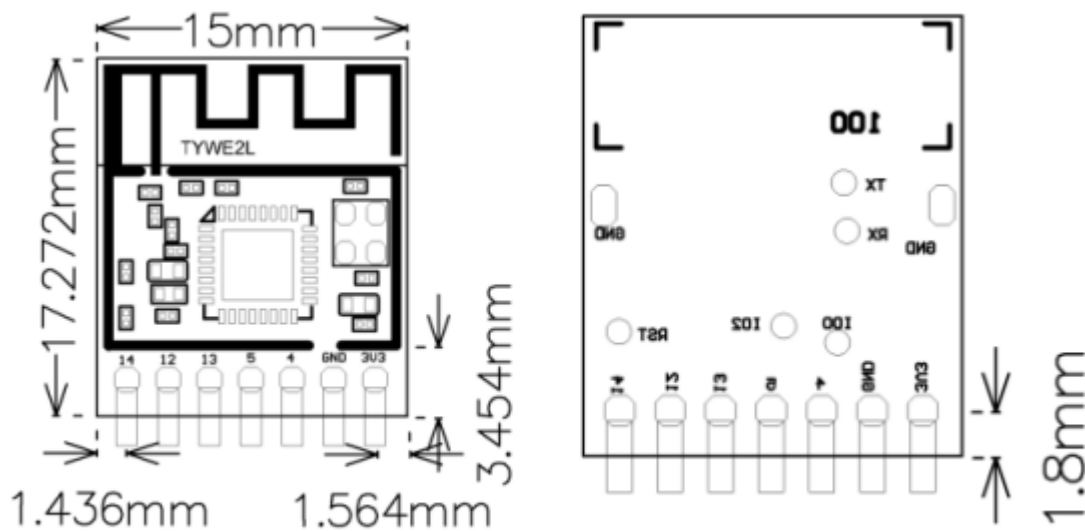


Figure 3 TYWE2L front view and rear view



Figure 4 TYWE2L side view

Note: PCB frame tolerance ± 0.15 mm, PCB depth tolerance ± 0.1 mm

6.2 Production instructions

Storage conditions of a delivered module are as follows:

1. The anti-moisture bag is placed in an environment where the temperature is under 30°C and the relative humidity is under 85%.

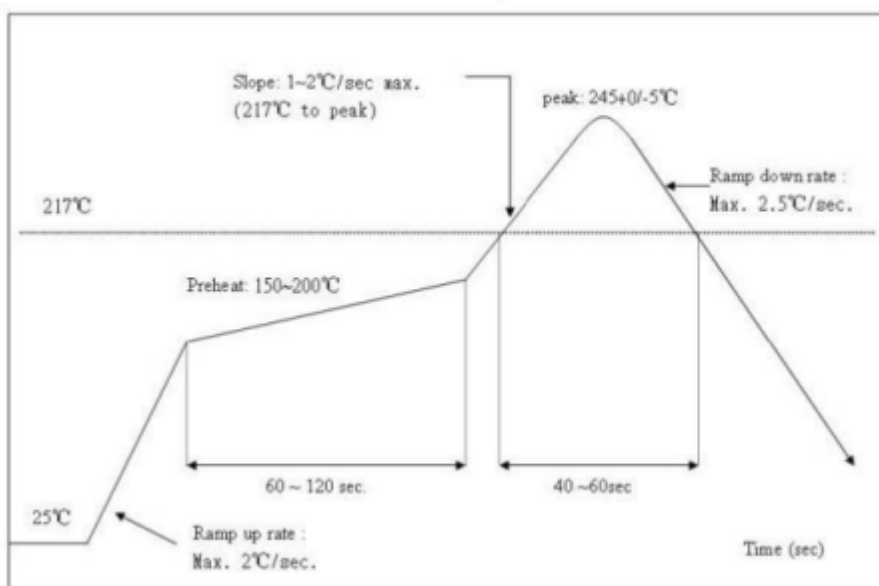
2. The shelf life of a dry-packaged product is six months from the date when the product is packaged and sealed.

Precautions:

1. Throughout the production process, each involved operator must wear an electrostatic ring.
2. During the operation, strictly protect the module from water and strains.

6.3 Recommended oven temperature profile

Refer to IPC/JEDEC standard ; Peak Temperature : $<250^{\circ}\text{C}$; Number of Times : ≤ 2 times ;



7 Appendix: Statement

FCC Caution: Any changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate this device.

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

Note: This device has been tested and found to comply with the limits for a Class B digital device, according to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This device generates, uses, and can radiate radio frequency energy and, if not installed and used following the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation.

If this device does cause harmful interference to radio or television reception, which can be determined by turning the device off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the device and receiver.
- Connect the device into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

Radiation Exposure Statement

This device complies with FCC radiation exposure limits set forth for an uncontrolled rolled environment. This device should be installed and operated with a minimum distance of 20cm between the radiator and your body.

Important Note

This radio module must not be installed to co-locate and operating simultaneously with other radios in the host system except following FCC multi-transmitter product procedures. Additional testing and device authorization may be required to operate simultaneously with other radios.

The availability of some specific channels and/or operational frequency bands are

country dependent and are firmware programmed at the factory to match the intended destination. The firmware setting is not accessible by the end-user.

The host product manufacturer is responsible for compliance with any other FCC rules that apply to the host not covered by the modular transmitter grant of certification. The final host product still requires Part 15 Subpart B compliance testing with the modular transmitter installed.

The end-user manual shall include all required regulatory information/warnings as shown in this manual, including “This product must be installed and operated with a minimum distance of 20 cm between the radiator and user body”.

This device has got an FCC ID: 2ANDL-TYWE2L. The end product must be labeled in a visible area with the following: “Contains Transmitter Module FCC ID: 2ANDL-TYWE2L”.

This device is intended only for OEM integrators under the following conditions:

The antenna must be installed such that 20cm is maintained between the antenna and users, and the transmitter module may not be co-located with any other transmitter or antenna.

As long as the 2 conditions above are met, further transmitter tests will not be required. However, the OEM integrator is still responsible for testing their end-product for any additional compliance requirements required with this module installed.

Declaration of Conformity European Notice



Hereby, Hangzhou Tuya Information Technology Co., Ltd declares that this module product is in compliance with essential requirements and other relevant provisions of Directive 2014/53/EU, 2011/65/EU. A copy of the Declaration of conformity can be found at <https://www.tuya.com>.



This product must not be disposed of as normal household waste, in accordance with the EU directive for waste electrical and electronic equipment (WEEE-2012/19/EU). Instead, it should be disposed of by returning it to the point of sale, or to a municipal recycling collection point.

The device could be used with a separation distance of 20cm to the human body.